

Example of course: Introduction to Metagenomic Next-Generation Sequencing:
Methods and Analysis for Pathogen Surveillance and Discovery

Course Overview and participant recommendations

Workshop Description:

The mNGS workshop will have two modules: laboratory theory and hands-on sessions, followed by no-code data analysis and interpretation. During the first week, participants will generate metagenomic libraries in the lab, gaining skills in Illumina library preparation, troubleshooting, and sequencing. Subsequently, the second week will center on the analysis and interpretation of metagenomic sequencing data utilizing the [CZ ID platform](#).

Ideal candidates for the workshop are those engaged in, or planning to undertake, research that could leverage Next Generation Sequencing (NGS). Candidates should possess foundational skills in molecular biology. For the second week, we encourage individuals engaged in mNGS and pathogen genomics data generation and/or analysis and interpretation to participate.

Workshop dates:

Workshop objective:

Prepare participants to undertake metagenomic NGS projects including study design, library preparation, and analysis and interpretation of results.

Week 1: Train wet lab scientists through theory sessions and hands-on laboratory sessions

Week 2: Learn to interpret results and analyze mNGS data using CZ ID.

Suggested previous experience of participants:

- Prioritize individuals who have projects in the near future that involve NGS and will continue to apply the hands-on skills.
- Week 1 + 2: Geared toward wet lab scientists who routinely complete hands-on molecular biology laboratory work
 - Required: proficient at pipetting techniques
 - Other: experience with PCR is highly recommended; experience with library preparation is not required
- Week 2: Focused on interpretation of results and analysis using a tool that does not require coding, appropriate for wet lab scientists and analysts

Week 1: Hands-on RNA Library Preparation and Sequencing

Day	Main Objectives	Lesson plan
Monday	• Introduce mNGS concepts • Define the role of mNGS in infectious diseases • Review study design • Clean, size select, and quantify DNA	• Lecture: Introduction to mNGS • Lecture and discussion: “mNGS in the context of infectious diseases” • Lecture: “Study design” • Hands-on: SPRI bead clean-up
Tuesday	• Understand the lab process to prepare Illumina libraries • Synthesize dsDNA for mNGS library preparation	• Lecture: “From sample to sequences” • Hands-on: Follow protocol for RNA fragmentation and dsDNA synthesis
Wednesday	• Ligate adapters to end repaired DNA fragments synthesized on day 2 • Understand the concepts underlying quality control of Illumina sequencing libraries	• Hands-on: Follow protocol for end repair, adapter ligation and barcoding PCR • Lecture: “Illumina library QC & pooling” • Hands-on: Follow protocol for library clean-up, size selection
Thursday	• Understand the principles underlying Illumina sequencing • Perform quality control on the libraries prepared during the training	• Lecture: “Illumina sequencing theory” • Hands-on: library quality control and pooling
Friday	• Load and sequence a library pool on a MiniSeq • Learn how to use Basespace and interpret sequencing quality parameters	• Hands-on: Library pool dilution and MiniSeq loading • Lecture: “Basespace and Sequencer Quality Control”

Week 2: Analysis and Interpretation of Sequencing Data Utilizing CZ ID and Other Bioinformatics Tools

Day	Main Objectives	Lesson plan
Monday	- Understand basic lab methods used for WGS and mNGS analysis - Understand key analysis steps in WGS and mNGS pipelines - Hands-on practice identifying sequences using BLAST	- Introductions & Syllabus Overview - Lecture: "Methods used in WGS and mNGS" - Lecture: "Bioinformatics basics" - Activity: BLAST
Tuesday	- Overview of different sequencing methods in the wet lab and implications for analysis - Introduction to the CZ ID platform - Understand the CZ ID mNGS pipeline and how to interpret results for a single sample in CZ ID - Hands-on practice interpreting mNGS results for a single sample in CZ ID - Discuss contamination issues and how we can address it in CZ ID	- Lecture: "Sequencing methods and platforms" - Lecture: Intro to CZ ID - Lecture: "CZ ID mNGS analysis" - Activity: Look at data! (example datasets) - Lecture: What about contamination?
Wednesday	- Learn how to use heatmaps in CZ ID to visualize taxa detected across samples - Hands-on practice interpreting heatmaps - Learn about mystery cases that you will need to solve with your group. Can you find a potential etiological agent based on the data?	- Lecture: Heatmap analysis - Activity: Heatmap analysis activity - Lecture: Mystery case intro: Are ready to solve a mystery case? - Activity: Mystery case assignment: Divide into groups and assign cases
Thursday	- Understand consensus genomes and assembly metrics to assess genome quality - Hands-on practice evaluating genome quality and completeness - Understand what is genomic epidemiology and what genomic information can tell us - Investigation of case studies	- Lecture: Assembling viral genomes in SeqtoID - Activity: Evaluate genome assembly - Lecture: Genomic epidemiology concepts & examples - Activity: Interpreting phylogenetic trees
Friday	- Guest lecture to enhance mNGS - AMR concepts and interpretation - Case Study presentations	- Use cases of mNGS in public health - Activity: AMR case study - Activity: all groups present case studies

Disclaimer:

- These training resources were developed to be used with the CZ ID platform.
- In October 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqToID (access at: <https://seqtoid.org/>).
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqToID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis.